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Class factorized complex variational auto-encoder for HRR radar target recognition

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ABSTRACT

In the field of radar automatic target recognition (RATR), the high-resolution range profile (HRRP) has received intensive attention. Bar a few exceptions, almost all HRRP-based ATR classification systems ignore the phase of the HRRP when the data is input to the classifier, relying instead only on the magnitude of the complex HRRP samples. This approach ignores the phase of the complex HRRPs, which reduces the information in the signal. In this paper, we develop a novel class factorized complex variational auto-encoder (CFCVAE) to utilize the phase of the high range resolution (HRR) radar echo for recognition. The CFCVAE is a complex-valued neural network (CV-NN) consisting of the encoding and decoding modules. In CFCVAE, the encoding module projects the observed data into the latent space, and then the latent features are fed to the decoding module, which further maps the latent features to data. In particular, the decoding module introduces the class labels to partition the whole observations into some parts, each of which is depicted by a specific class-decoder. Compared with the traditional variational auto-encoder (VAE) containing a single decoder, the CFCVAE can give a more accurate description to the whole dataset via multiple class-decoders, thus improving the characterization ability of features. In addition, based on the class labels, the CFCVAE employs the conditional prior on the latent variable to enhance the discrimination of features. Moreover, a complex backpropagation algorithm is derived for CFCVAE training, and a sample is classified to the class corresponding to the class-decoder with the minimum reconstruction error in the test stage. Experimental evaluations on the measured data indicate that the proposed method indeed achieves very promising target recognition performance.

1. Introduction

The high range resolution (HRR) radar usually works in the microwave band, and the size of the target is far larger than wavelength. The radar echo in the high-frequency region, i.e., the high range resolution (HRR) radar echo, can be well approximated as a sum of responses from individual scattering centers along the radar Line-Of-Sight (LOS). The HRR radar echo contains useful target structure characteristic, such as target size, scatterer distribution and so on. Moreover, since the HRR radar echo is easy to be collected and processed, the HRR radar target recognition has received extensive attention in the field of radar automatic target recognition (RATR) [1–10,17].

Most of the existing HRR radar target recognition methods directly utilize the amplitude of the HRR radar echo, which is the real-valued signal and is universally referred to as the high-resolution range profile (HRRP[1]). The radar HRRP recognition methods [1–10] can be divided into the traditional approaches [1–6] and deep-learning approaches [7–10].

For the traditional radar HRRP recognition approaches [1–6], some early work extracts features with physical meanings, such as Bispectra [1], power spectrum, FFT-magnitude, and high-order spectra [6], for recognition. More recently, some models [2–5] have been built to learn data-driven features of the HRRP. Feng et al. [3] utilize the K-Singularly Valuable Decomposition (K-SVD) algorithm to learn the sparse overcomplete representations of HRRPs. Du et al. [4] adopt the Principal Component Analysis (PCA) to convert the HRRP data into the feature subspace represented with some linear uncorrelated principal components. Besides, Du et al. [2,5] make use of the Factor Analysis (FA) model to characterize the HRRP data with several low-dimensional latent factors. These

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[1] The main basic steps of obtaining the HRRP are as follows: 1) transmitting the Chirp signal; 2) receiving returned signal; 3) dechirping; 4) matched filtering; 5) performing a modular operation for the HRR radar echo.

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traditional methods have shown their effectiveness in practical applications.

As the neural networks [18] attract more and more attention in various fields, some deep learning methods have been developed for radar HRRP recognition. Feng et al. [7] present a deep architecture named Stacked Corrective Auto-encoders (SCAE) for hierarchical feature learning of the HRRP, adopting the average profile as the correction term. Xu et al. [8] propose a Target-Aware Recurrent Attentional Network (TARAN) to explore the sequential relationship between the range cells and discover the target area in HRRP, which improves the discriminative ability of learned features. By incorporating labels into the variational auto-encoder [10], Du et al. [9] build the Factorized Discriminative Conditional Variational Auto-Encoder (FDCVAE) model to learn the probabilistic discriminative features of HRRPs. These deep learning models are real-valued neural networks, which perform well in radar HRRP recognition.

Nevertheless, the above radar HRRP recognition methods discard the phase in the HRR radar echo, thus restricting the recognition performance. In [21], the question “what information is available in the phase of a radar range profile?" is first posed by Rihaczek. Then Rigling et al. [22] address the above question and investigate the phase of the complex range profile. They adopt an electromagnetic model to depict phase scattering mechanisms, and experiments show that the phase in complex echo could improve model performance. Moreover, Hu et al. [17] develop two complex Gaussian statistical models, i.e., complex Adaptive Gaussian Classifier (CAGC) and complex FA (CFA), to make use of the phase in the complex radar echo for recognition. Additionally, some researchers study the phase of complex-valued synthetic aperture radar (SAR) images. El-Darymli et al. [23] build a statistical model to characterize the phase of the single-channel SAR image and introduce a set of 15 phase-based features, which perform well on the classification of the Moving and Stationary Target Acquisition and Recognition (MSTAR) dataset [23]. Besides, El-Darymli et al. [24] construct two holism-based frameworks to learn features of the complex-valued SAR imagery, and the first framework is solely based on the phase chip, while the second utilizes 1-D representations of the complex-valued SAR chip. Compared with the baseline model that discards the phase of the MSTAR dataset, the two frameworks [24] both gain higher classification accuracies. Furthermore, Coman et al. [25] analyze the potential of the additional radar information, such as the phase information, in the deep learning process. And experiments on the MSTAR dataset show that the phase is beneficial to increase the classification accuracy. Consequently, in the work [17,21–25], the discussion about the phase of the complex HRRPs and SAR images validates its usefulness for target recognition. More specifically, the phase of HRR radar echo contains target scatterers distribution, which reflects the target structure and scattering characteristics.

To take advantage of the phase in HRR radar echo, the complex-valued neural network (CV-NN) [11,16] can be developed for target recognition. Inspired by the effectiveness of the VAE [10] in describing data, we extend the traditional VAE to a CV-NN framework, in which all elements including the input-output layer, the hidden layer and activation function, are complex-valued. Moreover, the class labels are introduced and a novel class factorized complex variational auto-encoder (CFCVAE) is ultimately proposed for HRR radar target recognition. In detail, the CFCVAE is composed of the encoding and decoding modules. Especially, the decoding module utilizes the class labels to define multiple class-decoders, each of which specializes in learning a probability function to depict a part of observed data. It is noting that multiple class-decoders guide generating the observed data of different classes in the separable direction. In this way, our CFCVAE could depict all observations more precisely, thereof improving the expressiveness of the features to the whole dataset. In addition, we notice that the prior of latent variable determines how the latent space is structured. Therefore, by incorporating the class labels into the prior, our model adopts the conditional prior to enhance the separability of latent features. When training the CFCVAE model, a complex backpropagation algorithm is derived to jointly optimize the encoding and decoding modules. In the test stage, a sample is classified to the class that corresponds to the class-decoder with the minimum reconstruction error.

The main contributions of the research reported here are mainly as follows: 1) This paper focuses on the construction of a deep probability model named CFCVAE to exploit the phase of HRR radar echo for recognition. 2) By adopting multiple class-decoders, the CFCVAE is capable of learning the latent features with strong representation ability for the whole observations. 3) The CFCVAE employs the conditional prior based on the class labels, which enhances the discriminative ability of latent features.

The remainder of this paper is organized as follows. In Section 2, we present the analysis of HRR radar echo. Then, Section 3 gives a detailed introduction to the CFCVAE. In Section 4, experiments on the measured data are performed to evaluate the performance of our CFCVAE. Finally, the conclusions of the paper are summarized in Section 5.

2. The analysis of HRR radar echo

In this section, we focus on presenting a detailed introduction to the HRR radar echo, including the procedure of obtaining the signal and the analysis of the phase in the complex echo. The derivation of the HRR radar echo is helpful to obtain the expression of the signal, which is the input of our method. Moreover, in the HRR radar echo, the usefulness of phase motivates the development of our method to utilize the phase for recognition.

A standard processing chain of the HRR radar echo [19,29,30] is presented in Fig. 1. Combining with the flowchart, we will give a detailed derivation to the procedure of obtaining the HRR radar echo.

The transmitted signal $x_t(t)$ is the linear frequency modulated (LFM) signal,

$$\begin{aligned} x_t(t) &= s(t)e^{j2\pi f_c t} \\ &= rect\left(\frac{t}{T_p}\right)\exp\left(j\pi\mu t^2\right)\exp\left(j2\pi f_c t\right), \end{aligned} \tag{1}$$

where $s(\cdot)$ denotes the signal envelope, $rect(\frac{t}{T_p})$ denotes a rectangular pulse of the pulse width T_p, f_c represents the radar center frequency, μ represents the LFM coefficient with the bandwidth $B = \mu T_p$.

Then, the received signal is expressed as

$$x_r(t) = r(t) + c(t) + e(t), \tag{2}$$

where $r(t)$ denotes the target signal, $c(t)$ denotes the clutter, and $e(t)$ represents the noise. With the dechirping technique [28,29], the mth received signal in the nth range cell ($n = 1, 2, \cdots, N$) in baseband is denoted as

$$\hat{x}_n(t, m) = \sum_{i=1}^{L_n} \sigma_{ni} s\left(t - \frac{2R_{ni}(m)}{c}\right) e^{-j2\pi f_c \frac{2R_{ni}(m)}{c}} + c(t) + e(t), \tag{3}$$

where L_n represents the number of target scatterers in the nth range cell, σ_{ni} represents the magnitude of the ith scatterer in the nth range cell, $R_{ni}(m)$ denotes the radial distance between the radar and the ith scatterer of the nth range cell in the mth echo, c denotes the propagation velocity of radar signal. For a wideband radar system, the range dimension is divided into the range cells by the range resolution $\Delta R = \frac{c}{2B}$, which is the minimum range that radar can distinguish. Since the target scatterers in a range cell are

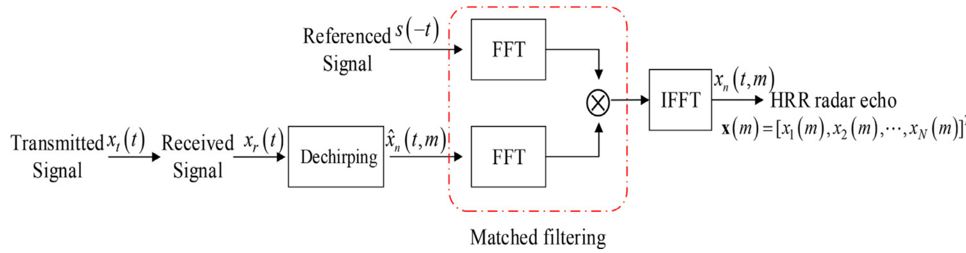

Fig. 1. The standard processing chain of obtaining the HRR radar echo.

approximately regarded a whole, the target signal of a range cell is the coherent summations of the complex time returns from the target scatterers in the range cell [19,27].

By matched filtering, we can get

$$\begin{aligned} F(x_n(t,m)) &= F\big(\hat{x}_n(t,m)\big)\cdot s^*(f) \\ &= \sum_{i=1}^{L_n}\sigma_{ni}s(f)e^{-j2\pi f\frac{2R_{ni}(m)}{c}}e^{-j2\pi f_c\frac{2R_{ni}(m)}{c}}s^*(f) \\ &\quad +C(f)+E(f) \\ &= \sum_{i=1}^{L_n}\sigma_{ni}P(f)e^{-j2\pi f\frac{2R_{ni}(m)}{c}}e^{-j2\pi f_c\frac{2R_{ni}(m)}{c}} \\ &\quad +C(f)+E(f), \end{aligned} \tag{4}$$

where $F(\cdot)$ denotes the Fourier transform; $s(f)$ represents the frequency response of $s(t)$; $s^*(f)$ denotes the conjugate form of $s(f)$; $C(f)=F(c(f))\cdot s^*(f)$, $N(f)=F(n(t))\cdot s^*(f)$; $P(f)$ represents the power spectrum of $s(t)$. For a LFM signal, $P(f)\approx rect(\frac{f}{B})$. When $-\frac{B}{2}<f<\frac{B}{2}$, Eq. (4) can be rewritten as

$$F(x_n(t,m)) = \sum_{i=1}^{L_n}\sigma_{ni}e^{-j2\pi f\frac{2R_{ni}(m)}{c}}e^{-j2\pi f_c\frac{2R_{ni}(m)}{c}}+C(f)+E(f). \tag{5}$$

After performing inverse Fourier transform (IFFT), the Eq. (5) is transformed to the time domain, and the received signal can be expressed as

$$\begin{aligned} x_n(t,m) &= \sum_{i=1}^{L_n}\sigma_{ni}e^{-j\frac{4\pi f_c}{c}R_{ni}(m)}\mathrm{sinc}\left(B\left(t-\frac{2R_{ni}(m)}{c}\right)\right) \\ &\quad +\hat{c}(t)+\hat{e}(t), \end{aligned} \tag{6}$$

where $\mathrm{sinc}(Bt)=\frac{\sin(\pi Bt)}{\pi Bt}$, $\hat{c}(t)=F^{-1}(C(f))$, $\hat{e}(t)=F^{-1}(E(f))$ with $F^{-1}(\cdot)$ denoting IFFT operation.

Moreover, Eq. (6) can be further expended as

$$\begin{aligned} x_n(t,m) &= \sum_{i=1}^{L_n}\sigma_{ni}e^{-j\frac{4\pi f_c}{c}(R(m)+r_{ni}(m))}\mathrm{sinc}\left(B\left(t-\frac{2R_{ni}(m)}{c}\right)\right) \\ &\quad +\hat{c}(t)+\hat{e}(t) \\ &= e^{j\psi(m)}\sum_{i=1}^{L_n}\sigma_{ni}e^{j\phi_{ni}(m)}\mathrm{sinc}\left(B\left(t-\frac{2R_{ni}(m)}{c}\right)\right) \\ &\quad +\hat{c}(t)+\hat{e}(t), \end{aligned} \tag{7}$$

where $R(m)$ denotes the radial distance between the target reference center in the mth returned echo and the radar, $r_{ni}(m)$ represents the radial distance of the ith scatterer of the nth range cell in the mth returned echo to the center of the target, and $\psi(m)=-j\frac{4\pi f_c}{c}R(m)$ denotes the initial phase of the mth returned echo, $\phi_{ni}(m)=-j\frac{4\pi f_c}{c}r_{ni}(m)$ denotes the phase of the ith scatterer of the nth range cell in mth returned echo. Since the radial displacement of all scatterers in the nth range resolution cell is approximately unvaried, the mth HRR radar echo in the nth range cell can be approximated as

$$\begin{aligned} x_n(m) &\approx e^{j\psi(m)}\sum_{i=1}^{L_n}\sigma_{ni}e^{j\phi_{ni}(m)}+\hat{c}_n(m)+\hat{e}_n(m) \\ &= e^{j\psi(m)}\gamma_n(m)e^{j\varphi_n(m)}+\hat{c}_n(m)+\hat{e}_n(m), \end{aligned} \tag{8}$$

where $\gamma_n(m)$ denotes the amplitude of nth range cell of the mth returned echo, $\varphi_n(m)$ denotes the phase of the nth range cell of the mth returned echo. As we can see from Eq. (8), the phase of the mth radar echo contains two parts: one part $\psi(m)=-\frac{4\pi}{\lambda}R(m)$ is only related to the radial distance $R(m)$, which is independent of the target structure; another part $\varphi_n(m)$ can be further expressed as

$$\varphi_n(m) = \arctan\left(\frac{\sum_{i=1}^{L_n}\sigma_{ni}\sin\left(-\frac{4\pi f_c}{c}r_{ni}(m)\right)}{\sum_{i=1}^{L_n}\sigma_{ni}\cos\left(-\frac{4\pi f_c}{c}r_{ni}(m)\right)}\right) \tag{9}$$

with $\arctan(\cdot)$ denoting the inverse tangent function. As shown in Eq. (9), $\varphi_n(m)$ contains the magnitude and locations of all of the target scatterers in the nth range cell of the mth returned echo. Obviously the target scatterer distribution is embedded in the phase $\varphi_n(m)$, which is valuable for recognition.

As for the mth HRR radar echo, it is defined as

$$\mathbf{x}(m) = [x_1(m), x_2(m), \cdots, x_N(m)]^T, \tag{10}$$

where T represents the transpose operation. The amplitude of the mth HRR radar echo, i.e., high-resolution range profile (HRRP), can be expressed as

$$\boldsymbol{\gamma}(m) = |\mathbf{x}(m)| = [|x_1(m)|, |x_2(m)|, \cdots, |x_N(m)|]^T \tag{11}$$

with $|\cdot|$ denoting modular arithmetic operation.

As shown in Eq. (8) and Eq. (10), the HRR radar echo contains the clutter and noise. Generally, in the radar signal processing system, clutter suppression [31–33] and target detection [34,35] are first applied to the HRR radar echo, and then the processed signal is further utilized for target recognition. Thus, the impact of the clutter can be ignored in target recognition. As for the noise in the HRR radar echo, the most direct way is to utilize some noise reduction methods, such as the orthogonal matching pursuit (OMP) algorithm [20]. While in the test stage of recognition, the denoising processing of a sample is time-consuming in the real-time decision. Therefore, it is better to possess the noise robustness characteristic for a recognition method. This paper proposes a method that is robust to the noise by learning noise-robust features for recognition.

3. The proposed method

Since the phase of HRR radar echo contains the target structure characteristic, we propose a complex-valued neural network (CV-NN), called class factorized complex variational auto-encoder (CFC-VAE), to utilize the phase of HRR radar echo for target recognition. In Section 3.1, we introduce the model structure of CFCVAE. And in Section 3.2, we present the optimization solution and prediction of the proposed model, covering the model training and test procedures.

3.1. Model structure

We posit that the whole dataset $\mathbf{X}=\{\mathbf{x}_n\}_{n=1}^N$ contains N samples covering C classes and $\mathbf{x}_n \in \mathbb{C}^{P\times 1}$ with P denoting the dimension of $\mathbf{x}_n$. The class labels of all samples are represented as $\mathbf{Y}=\{\mathbf{y}_n\}_{n=1}^N$ with $\mathbf{y}_n \in \mathbb{R}^{C\times 1}$ being a one-hot vector. Thus, the decoding module of the CFCVAE contains C class-decoders, in which the cth class-decoder with parameter θ_{c_n} depicts the generative scheme from the latent variable $\mathbf{z}_n$ to a sample $\mathbf{x}_n$ via $p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)$ with $c_n \in \{1,\cdots,C\}$ denoting the indictor of nth sample and $\mathbf{c}=\{c_n\}_{n=1}^N$. In CFCVAE, we adopt the solving trick used in the traditional VAE [10] for finding the estimation of $\boldsymbol{\theta}$. Therefore, we have

$$\begin{aligned}\ln p_{\boldsymbol{\theta}}(\mathbf{X}|\mathbf{c}) &= \sum_{n=1}^{N}\ln\left(\int p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)p(\mathbf{z}_n)d\mathbf{z}_n\right)\\ &\geq \sum_{n=1}^{N}\mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)}\left[\ln p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)\right]\\ &\quad -\mathrm{KL}\big(q_\varphi(\mathbf{z}_n|\mathbf{x}_n)\|p(\mathbf{z}_n)\big)\\ &= \mathrm{LB}\big(\boldsymbol{\theta},\varphi;\mathbf{X},\mathbf{c}\big), \end{aligned} \tag{12}$$

where $\mathrm{LB}(\boldsymbol{\theta},\varphi;\mathbf{X},\mathbf{c})$ denotes the (variational) lower bound on the marginal likelihood, the expectation of log likelihood term $\mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)}[\ln p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)]$ indicates that each sample $\mathbf{x}_n$ $(n=1,\cdots,N)$ is generated from the latent variable $\mathbf{z}_n$ by a specified class-decoder with parameter θ_{c_n}, the KL divergence term $\mathrm{KL}(q_\varphi(\mathbf{z}_n|\mathbf{x}_n)\|p(\mathbf{z}_n))$ encourages the approximate posterior close to the prior distribution. Moreover, based on the label $\mathbf{y}_n$, the CFCVAE employs the conditional prior $p_\phi(\mathbf{z}_n|\mathbf{y}_n)$ to help learn the discriminative features. The lower bound can be further expanded as

$$\begin{aligned}\mathrm{LB}\big(\boldsymbol{\theta},\varphi,\phi;\mathbf{X},\mathbf{c}\big) &= \sum_{n=1}^{N}\mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)}\left[\ln p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)\right]\\ &\quad -\mathrm{KL}\big(q_\varphi(\mathbf{z}_n|\mathbf{x}_n)\big\|p_\phi(\mathbf{z}_n|\mathbf{y}_n)\big), \end{aligned} \tag{13}$$

where ϕ denotes the network parameter that is used to infer the prior of the latent variable $\mathbf{z}_n$. According to Eq. (13), the optimization of the parameter θ_c $(c=1,\cdots,C)$ just depends on those samples belonging to the cth class, while all samples contribute to the learning of the network parameter φ, and the class labels devote to the learning of the network parameter ϕ.

In Eq. (13), $\mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)}[\ln p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)]$ can be calculated via the Monte Carlo estimator with $\frac{1}{L}\sum_{l=1}^{L}\ln p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n^l)$, where L represents the number of samples and $\mathbf{z}_n^l$ denotes the lth sampling value of $\mathbf{z}_n$ sampled from $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$. Nevertheless, the sampling of $\mathbf{z}_n$ hinders the back-propagation of error. To address the problem, the reparameterization trick [10] is employed by introducing an auxiliary noise variable $\boldsymbol{\varepsilon}_n$ to represent $\mathbf{z}_n$ as $\mathbf{z}_n=g_\varphi(\mathbf{x}_n,\boldsymbol{\varepsilon}_n)$ with $\boldsymbol{\varepsilon}_n \sim p(\boldsymbol{\varepsilon}_n)$.

Specifically, in CFCVAE, we take the multivariate complex Gaussian distribution as the posterior of the complex-valued variable $\mathbf{z}_n$ with its mean $\boldsymbol{\mu}_\varphi(\mathbf{x}_n)$ and standard deviation $\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)$, and the posterior is expressed as

$$q_\varphi(\mathbf{z}_n|\mathbf{x}_n)=\mathbb{CN}\Big(\mathbf{z}_n\Big|\boldsymbol{\mu}_\varphi(\mathbf{x}_n),diag\big(\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)\big)\Big), \tag{14}$$

where $\mathbb{CN}(\cdot)$ denotes the complex Gaussian distribution, and diag$(\cdot)$ represents the diagonalization. Based on the reparameterization trick, the variable $\mathbf{z}_n$ can be represented as $\mathbf{z}_n=\mu\varphi(\mathbf{x}_n)+\sigma_\varphi(\mathbf{x}_n)\odot\varepsilon_n,\ \ \varepsilon_n\sim\mathbb{CN}(\mathbf{0},\mathbf{I})$. Likewise, the conditional prior $p_\phi(\mathbf{z}_n|\mathbf{y}_n)$ is also assumed to be the multivariate complex Gaussian distribution with the form of

$$p_\phi(\mathbf{z}_n|\mathbf{y}_n)=\mathbb{CN}\Big(\mathbf{z}_n\Big|\boldsymbol{\mu}_\phi(\mathbf{y}_n),diag\big(\boldsymbol{\sigma}_\phi(\mathbf{y}_n)\big)\Big), \tag{15}$$

where $\boldsymbol{\mu}_\phi(\mathbf{y}_n)$ and $\boldsymbol{\sigma}_\phi(\mathbf{y}_n)$ denote the mean and standard deviation, respectively. Therefore, the KL divergence[2] between two complex Gaussian distributions is explicitly calculated as

$$\begin{aligned}&\mathrm{KL}\big(q_\varphi(\mathbf{z}_n|\mathbf{x}_n)\big\|p_\phi(\mathbf{z}_n|\mathbf{y}_n)\big)\\ &=-\textstyle\sum_{p=1}^{P}\left[1+\ln\left(\frac{\sigma_\varphi^{(n,p)}}{\sigma_\phi^{(n,p)}}\right)^2-\left(\frac{\sigma_\varphi^{(n,p)}}{\sigma_\phi^{(n,p)}}\right)^2-\left(\frac{\mu_\varphi^{(n,p)}-\mu_\phi^{(n,p)}}{\sigma_\phi^{(n,p)}}\right)^2\right], \end{aligned} \tag{16}$$

where $\mu_\varphi^{(n,p)}$ and $\mu_\phi^{(n,p)}$ $(p=1,\cdots,P)$ signify the pth element of $\boldsymbol{\mu}_\varphi(\mathbf{x}_n)$ and $\boldsymbol{\mu}_\phi(\mathbf{y}_n)$, respectively, and similar meaning for $\sigma_\varphi^{(n,p)}$, $\sigma_\phi^{(n,p)}$.

Combining with the Eq. (13), (14), (15) and (16), the eventual estimator of the CFCVAE could be rewritten as

$$\begin{aligned}&\mathrm{LB}\big(\boldsymbol{\theta},\varphi,\phi;\mathbf{X},\mathbf{c}\big)\simeq\frac{1}{L}\sum_{n=1}^{N}\sum_{l=1}^{L}\ln p_{\theta_{c_n}}\big(\mathbf{x}_n\big|\mathbf{z}_n^l\big)\\ &-\sum_{n=1}^{N}\sum_{p=1}^{P}\left[1+\ln\left(\frac{\sigma_\varphi^{(n,p)}}{\sigma_\phi^{(n,p)}}\right)^2-\left(\frac{\sigma_\varphi^{(n,p)}}{\sigma_\phi^{(n,p)}}\right)^2-\left(\frac{\mu_\varphi^{(n,p)}-\mu_\phi^{(n,p)}}{\sigma_\phi^{(n,p)}}\right)^2\right]\end{aligned}$$

with

$$\mathbf{z}_n^l=\boldsymbol{\mu}_\varphi(\mathbf{x}_n)+\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)\odot\boldsymbol{\varepsilon}_n^l,\ \boldsymbol{\varepsilon}_n^l\sim\mathbb{CN}(\mathbf{0},\mathbf{I}), \tag{17}$$

where $\boldsymbol{\varepsilon}_n^l$ denotes the lth sampling value of $\boldsymbol{\varepsilon}_n$. For the intuitive illustration, the whole network architecture of the CFCVAE is shown in Fig. 2.

3.2. Optimization solution and prediction

3.2.1. Training procedure

In the model training stage, we first deduce the feed-forward propagation of the CFCVAE, and then a complex backpropagation algorithm [12] is applied to obtain the solutions of parameters $\boldsymbol{\theta}$, φ and ϕ.

We adopt multilayer perception (MLP) architecture to achieve strong expression ability of our CFCVAE model, which means that the proposed model can learn the deep features characterizing the essential property of the observed data. In CFCVAE, the layer numbers of the MLP should be adjusted, depending on the data type. Here, the MLP with a single hidden layer is taken as an example to present the derivations of the latent variable $\mathbf{z}$ and the corresponding observed data $\mathbf{x}$.

In detail, the conditional prior of $\mathbf{z}$, i.e., $p_\phi(\mathbf{z}|\mathbf{y})=\mathbb{CN}(\mathbf{z}|\boldsymbol{\mu}_\phi(\mathbf{y}),diag(\boldsymbol{\sigma}_\phi(\mathbf{y})))$, is derived by the network with parameter ϕ:

$$\mathbf{h}_1=\mathrm{sigm}(\Re(\mathbf{W}_1\mathbf{y}+\mathbf{b}_1))+j\mathrm{sigm}(\Im(\mathbf{W}_1\mathbf{y}+\mathbf{b}_1)), \tag{18}$$

$$\boldsymbol{\mu}_\phi(\mathbf{y})=\mathbf{W}_2\mathbf{h}_1+\mathbf{b}_2, \tag{19}$$

$$\hat{\sigma}_\phi(\mathbf{y})=\mathrm{sigm}(\Re(\mathbf{W}_3\mathbf{h}_1+\mathbf{b}_3))+j\mathrm{sigm}(\Im(\mathbf{W}_3\mathbf{h}_1+\mathbf{b}_3))$$

[2] The prior and posterior of the complex-valued latent variable $\mathbf{z}_n$ are both complex Gaussian distributions, represented with $p_\phi(\mathbf{z}_n|\mathbf{y}_n)=\mathbb{CN}(\mathbf{z}_n|\boldsymbol{\mu}_\phi(\mathbf{y}_n),diag(\boldsymbol{\sigma}_\phi(\mathbf{y}_n)))$ and $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)=\mathbb{CN}(\mathbf{z}_n|\boldsymbol{\mu}_\varphi(\mathbf{x}_n),diag(\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)))$, respectively. The KL divergence between $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$ and $p_\phi(\mathbf{z}_n|\mathbf{y}_n)$ can be expanded as

$$\begin{aligned}&\mathrm{KL}(q_\varphi(\mathbf{z}_n|\mathbf{x}_n)\|p_\theta(\mathbf{z}_n))=\textstyle\int q_\varphi(\mathbf{z}_n|\mathbf{x}_n)\ln\frac{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)}{p_\theta(\mathbf{z}_n)}d\mathbf{z}_n\\ &=\textstyle\int q_\varphi(\mathbf{z}_n|\mathbf{x}_n)\ln\mathbb{CN}(\mathbf{z}_n|\boldsymbol{\mu}_\varphi(\mathbf{x}_n),\mathrm{diag}(\boldsymbol{\sigma}_\varphi^2(\mathbf{x}_n)))d\mathbf{z}_n\\ &\quad-\textstyle\int q_\varphi(\mathbf{z}_n|\mathbf{x}_n)\ln\mathbb{CN}(\mathbf{z}_n|\boldsymbol{\mu}_\phi(\mathbf{y}_n),diag(\boldsymbol{\sigma}_\phi(\mathbf{y}_n)))d\mathbf{z}_n\\ &=-\textstyle\sum_{p=1}^{P}[1+\ln(\frac{\sigma_\varphi^{(n,p)}}{\sigma_\phi^{(n,p)}})^2-(\frac{\sigma_\varphi^{(n,p)}}{\sigma_\phi^{(n,p)}})^2-(\frac{\mu_\varphi^{(n,p)}-\mu_\phi^{(n,p)}}{\sigma_\phi^{(n,p)}})^2]\end{aligned}$$

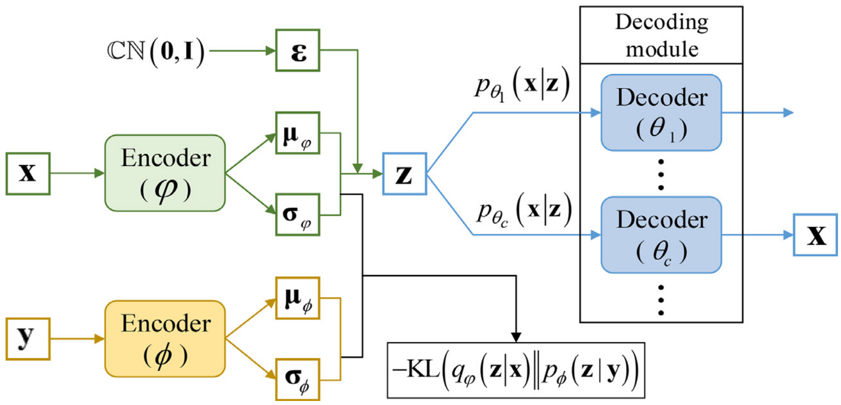

Fig. 2. Architecture of the CFCVAE model. The encoder with parameter φ projects the sample $\mathbf{x}$ to the latent space, and then the latent variable $\mathbf{z}$ is fed to the decoder with parameter θ_c based on the indictor c. The prior of the latent variable $\mathbf{z}$ is inferred via a network with parameter ϕ based on the label $\mathbf{y}$. The KL term pushes the posterior $q_\varphi(\mathbf{z}|\mathbf{x})$ close to the prior $p_\phi(\mathbf{z}|\mathbf{y})$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Algorithm 1 The training stage of the CFCVAE model.

Set the architecture of the CFCVAE model including the dimension of features and the number of hidden layers;
Initialize the network parameters ϕ, φ and $\boldsymbol{\theta}$;
while not converged **do**
  Randomly sample a mini-batch $\mathbf{X}_1 = \{\mathbf{x}_n\}_{n=1}^{N_1}$ from the whole dataset $\mathbf{X}$, then find the label $\mathbf{Y}_1 = \{\mathbf{y}_n\}_{n=1}^{N_1}$ and the indictor $\mathbf{c}_1 = \{c_n\}_{n=1}^{N_1}$;
  Sample random noise $\{\boldsymbol{\varepsilon}_n\}_{n=1}^{N_1}$ from standard complex Gaussian distribution for reparameterization;
  Get the prior of the latent variable $\mathbf{z}_n$ from $p_\phi(\mathbf{z}_n|\mathbf{y}_n)$ and the posterior from $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$, and then generate observation $\mathbf{x}_n$ from $p_{\theta_{c_n}}(\mathbf{x}_n|\mathbf{z}_n)$;
  Update ϕ, φ and $\boldsymbol{\theta}$ via SGA on the lower bound, i.e., Eq. (17).
end while

with

$$\boldsymbol{\sigma}_\phi(\mathbf{y}) = \left|\hat{\sigma}_\phi(\mathbf{y})\right|, \tag{20}$$

where $j=\sqrt{-1}$ denotes the imaginary unit, $\Re(\cdot)$ and $\Im(\cdot)$ represent the real and imaginary parts, respectively; $\phi = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \mathbf{W}_3, \mathbf{b}_3\}$ represent the network parameters with $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{W}_3$ denoting the weight matrices and $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$ the biases, respectively; $\mathrm{sigm}(\cdot)$ denotes the sigmoid function; $|\cdot|$ is modular arithmetic operation; $\mathbf{h}_1$ denotes the output of the hidden layer in the prior learning network. In Eq. (20), the term $\hat{\sigma}_\phi(\mathbf{y})$ obtained by the network parameterized with ϕ is a complex-valued variable. Since the standard deviation of a complex Gaussian distribution is real-valued, here we define $|\hat{\sigma}_\phi(\mathbf{y})|$ as the standard deviation.

Analogously, the posterior of $\mathbf{z}$, i.e., $q_\varphi(\mathbf{z}|\mathbf{x}) = \mathbb{CN}(\mathbf{z}|\boldsymbol{\mu}_\varphi(\mathbf{x}), diag(\boldsymbol{\sigma}_\varphi(\mathbf{x})))$, is inferred via the network parameterized by φ:

$$\mathbf{h}_2 = \mathrm{sigm}(\Re(\mathbf{W}_4\mathbf{x}+\mathbf{b}_4)) + j\mathrm{sigm}(\Im(\mathbf{W}_4\mathbf{x}+\mathbf{b}_4)), \tag{21}$$

$$\boldsymbol{\mu}_\varphi(\mathbf{x}) = \mathbf{W}_5\mathbf{h}_2 + \mathbf{b}_5, \tag{22}$$

$$\hat{\sigma}_\varphi(\mathbf{x}) = \mathrm{sigm}(\Re(\mathbf{W}_6\mathbf{h}_2+\mathbf{b}_6)) + j\mathrm{sigm}(\Im(\mathbf{W}_6\mathbf{h}_2+\mathbf{b}_6))$$

with

$$\boldsymbol{\sigma}_\varphi(\mathbf{x}) = \left|\hat{\sigma}_\varphi(\mathbf{x})\right|, \tag{23}$$

where $\varphi = \{\mathbf{W}_4, \mathbf{b}_4, \mathbf{W}_5, \mathbf{b}_5, \mathbf{W}_6, \mathbf{b}_6\}$ denote the network parameters.

In the decoding module, as for the observed sample $\mathbf{x}$ with the indicator c, it is obtained via the cth class-decoder. Similarly to the posterior of the variable $\mathbf{z}$, the conditional distribution of $\mathbf{x}$ is assumed to follow the complex Gaussian distribution, i.e., $p_{\theta_c}(\mathbf{x}|\mathbf{z}) = \mathbb{CN}(\mathbf{x}|\boldsymbol{\mu}_{\theta_c}(\mathbf{z}), diag(\boldsymbol{\sigma}_{\theta_c}(\mathbf{z})))$, and is inferred by

$$\mathbf{h}_3 = \mathrm{sigm}(\Re(\mathbf{W}_7\mathbf{z}+\mathbf{b}_7)) + j\mathrm{sigm}(\Im(\mathbf{W}_7\mathbf{z}+\mathbf{b}_7)), \tag{24}$$

$$\boldsymbol{\mu}_{\theta_c}(\mathbf{z}) = \mathbf{W}_8\mathbf{h}_3 + \mathbf{b}_8, \tag{25}$$

$$\hat{\sigma}_{\theta_c}(\mathbf{z}) = \mathrm{sigm}(\Re(\mathbf{W}_9\mathbf{h}_3+\mathbf{b}_9)) + j\mathrm{sigm}(\Im(\mathbf{W}_9\mathbf{h}_3+\mathbf{b}_9))$$

with

$$\boldsymbol{\sigma}_{\theta_c}(\mathbf{z}) = \left|\hat{\sigma}_{\theta_c}(\mathbf{z})\right|, \tag{26}$$

where $\theta_c = \{\mathbf{W}_7, \mathbf{b}_7, \mathbf{W}_8, \mathbf{b}_8, \mathbf{W}_9, \mathbf{b}_9\}$ denote the network parameters of the cth class-decoder.

Then, a complex backpropagation algorithm [12] based on stochastic gradient ascent (SGA) is derived to maximize the lower bound $\mathrm{LB}(\boldsymbol{\theta}, \varphi, \phi; \mathbf{X}, \mathbf{c})$ w.r.t. the network parameters ϕ, φ and $\boldsymbol{\theta}$. The total training process of the CFCVAE is outlined in Algorithm 1. The joint optimization of the encoding and decoding modules devotes to the superior performance of the CFCVAE.

3.2.2. Test procedure

After the model training process, the learned CFCVAE model can be used for the class prediction of test samples. For a complex-valued test sample $\mathbf{x}^*$, we can obtain the latent feature $\mathbf{z}^*$ by using the feed-forward propagation via the trained encoder $q_\varphi(\mathbf{z}^*|\mathbf{x}^*)$. The latent feature $\mathbf{z}^*$ is then fed to the C class-decoders $p_{\boldsymbol{\theta}}(\mathbf{x}^*|\mathbf{z}^*)$ with $\boldsymbol{\theta} = \{\theta_1, \cdots, \theta_c \cdots, \theta_C\}$ to generate a set of reconstructed samples $\{\hat{\mathbf{x}}_1^*, \cdots, \hat{\mathbf{x}}_c^*, \cdots, \hat{\mathbf{x}}_C^*\}$. The reconstruction errors between $\mathbf{x}^*$ and $\{\hat{\mathbf{x}}_1^*, \cdots, \hat{\mathbf{x}}_c^*, \cdots, \hat{\mathbf{x}}_C^*\}$ can be further calculated. Based on the minimum reconstruction error criterion, the test sample is classified to the c^*th class with

$$\begin{aligned} c^* &= \underset{c\in\{1,\cdots,C\}}{\arg\min} \left\|\mathbf{x}^* - \hat{\mathbf{x}}_c^*\right\|^2 \\ &= \underset{c\in\{1,\cdots,C\}}{\arg\min} \left\| |\mathbf{x}^*| e^{j\alpha} - \left|\hat{\mathbf{x}}_c^*\right| e^{j\hat{\alpha}_c^*} \right\|^2 \\ &= \underset{c\in\{1,\cdots,C\}}{\arg\min} \left(|\mathbf{x}^*|\cos\alpha - \left|\hat{\mathbf{x}}_c^*\right|\cos\hat{\alpha}_c^*\right)^2 \\ &\quad + \left(|\mathbf{x}^*|\sin\alpha - \left|\hat{\mathbf{x}}_c^*\right|\sin\hat{\alpha}_c^*\right)^2, \end{aligned} \tag{27}$$

Algorithm 2 The test stage of the CFCVAE model.

```
Input the test data x* into the trained encoder with parameter φ to obtain the latent feature z* via q_φ(z*|x*);
for c = 1 : C
    Fed the latent feature z* into the cth class-decoder with parameter θ_c, and obtain the output x̂*_c via p_{θ_c}(x*|z*);
    Calculate the reconstruction error corresponding to the cth class-decoder error_c = ‖ x* − x̂*_c ‖².
end for
According to the minimum reconstruction error criterion, obtain the index c* via Eq. (27), and then classify the test data x* to the c*th class.
```

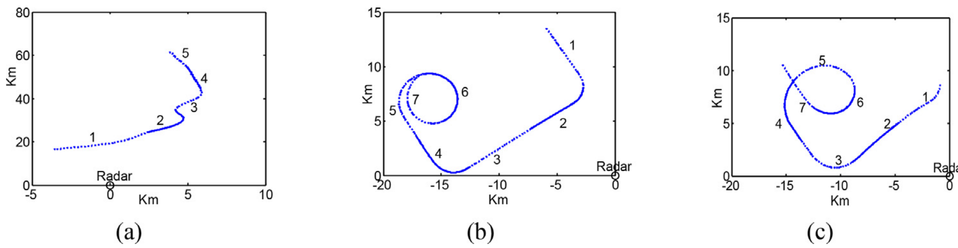

Fig. 3. Two-dimensional track charts of three radar targets. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

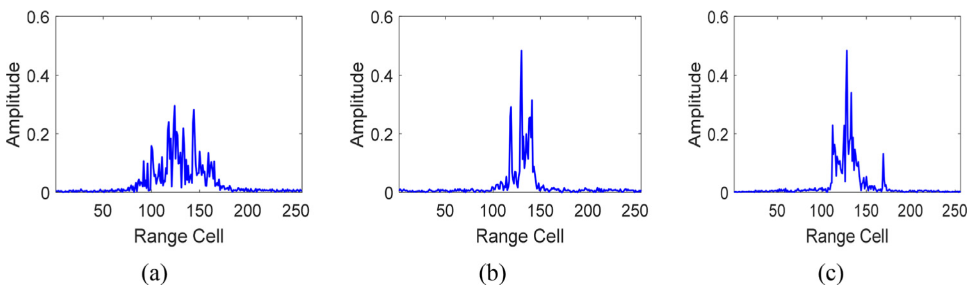

Fig. 4. The HRRPs of three radar targets. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

where $|\cdot|$ represents the modular arithmetic operation, α^* and $\hat{\alpha}_c^*$ denote the phase of $\mathbf{x}^*$ and $\hat{\mathbf{x}}_c^*$, respectively. As shown in Eq. (27), the phase is utilized to obtain the reconstruction error, which means that the phase of a complex-valued test sample is also utilized for recognition. The test process is presented in Algorithm 2. In addition, our method has much higher efficiency than some probabilistic models [2,5,15], which often adopt the variational Bayesian (VB) algorithm for iterative parameter inference at testing.

Table 1
Parameters of three airplanes and radar in the experiment.

Radar parameters	Transmitted signal waveform		Chirp signal
	Center frequency		5520 *MHZ*
	Bandwidth		400 *MHZ*
	Pulse repetition frequency		400 *HZ*
	Sampling frequency after dechirping		10 *MHZ*
Airplanes	Length (m)	Width(m)	Height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

4. Experiments

In this section, we comprehensively conduct experiments on the measured data to evaluate the performance of the proposed CFCVAE. In Section 4.1, we provide the introduction to the measured dataset and experimental settings, followed by the analysis of model parameters in Section 4.2. Then Section 4.3 and 4.4 respectively analyze the recognition performance and noise-robust characteristic of our model. Ultimately, the high time-efficiency of the proposed CFCVAE is validated in Section 4.5.

4.1. Data description and experimental setup

The radar dataset used in this paper is the aircraft data measured by ISAR radar of a domestic research institute, which is also used in [3–8,14,19]. The radar works in C-band and transmits chirp signal. There are three real airplanes, in which An-26 is a medium-sized propeller aircraft, Cessna Citation is a small-sized jet aircraft and Yark-42 is a large-sized jet aircraft. The parameters of the targets and radar are shown in Table 1. The projections of target trajectories onto the ground plane are shown in Fig. 3, from which the aspect angle of airplane can be estimated according to its relative position to radar. The measured data are segmented into several parts. The amplitude of the HRR radar echo is defined as the radar high-resolution range profile (HRRP). In Fig. 4, we present the HRRPs of the three aircraft, and the dimension of the HRRP is 256, i.e., $P = 256$. In addition, the radar dataset is obtained via the cooperative measurement experiments, which are under the condition of high signal-to-noise ratio ($SNR \approx 40dB$) and without any interference. Thus, the impact of the noise and clutter on recognition can be ignored.

We select the training and test samples according to the following preconditions: a) the target-aspect of the training samples should cover almost all those of the test samples; b) the elevation angles of the training and test samples should be different. Based on the requirements above, we select data from the 2nd and 5th segments of Yark-42, the 6th and 7th segments of Cessna Citation S/II, the 5th and 6th segments of An-26 as the training samples, and the data from other segments are taken as test samples. In the following experiments, the training and test datasets contain 138,240 and 5200 samples, respectively.

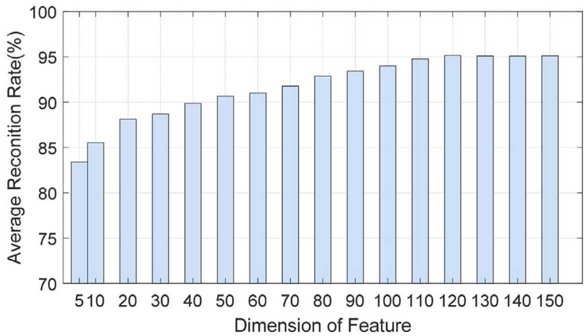

Fig. 5. Recognition performance of the CFCVAE with different dimensions of features.

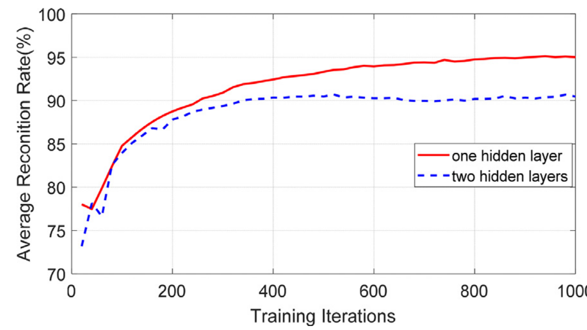

Fig. 6. Recognition performance of the CFCVAE with different numbers of hidden layers.

Since the measured dataset covers three classes, the CFCVAE model has three class-decoders, i.e., $C = 3$. The encoding architecture of our proposed model is set as 256−200−120, and the architecture of the prior learning network is set as 3−50−120. The real and imaginary parts of the network weights are initialized by sampling randomly from $\mathbb{N}(0, 1)$, respectively; and the bias terms are initialized as 0. We set the batchsize N_1 of training data as 100. The learning rate ρ of the SGA algorithm is set as 0.001. All the following experiments are performed in non-optimized software written in Matlab, on a Dell PC with 3.40 GHz CPU and 16GB RAM.

Moreover, we compare the CFCVAE with some related models, including some traditional shallow models and deep networks. The traditional shallow models: 1) Principal Component Analysis (PCA) [4], 2) K singular value decomposition (K-SVD) [3], 3) Adaptive Gaussian classifier (AGC) [13,17], 4) Factor Analysis (FA) [14,17], 5) Complex Adaptive Gaussian classifier (CAGC) [17], 6) Complex Factor Analysis (CFA) [17]. The deep networks: 1) Supervised VAE (SVAE), 2) Stacked Corrective Auto-encoder (SCAE) [7], 3) Target-Aware Recurrent Attentional Network (TARAN) [8], 4) Factorized Discriminative Conditional Variational Auto-encoder (FDCVAE) [9], 5) Stacked Denoising Auto-encoder (SDAE) [12].

4.2. The analysis of model parameters

To choose the befitting network architecture of the CFCVAE, the effect of model parameters on the recognition performance is analyzed. The model parameters include the dimension of features and the number of hidden layers.

Fig. 5 depicts the recognition performance of our model with different dimensions of features. The larger the number of hidden units is, the more representative features are learned. Thus the model with more hidden units could depict the observations better and gain better recognition performance. In Fig. 5, as the dimension of the hidden feature increases from 5 to 120, the recognition performance of the CFCVAE improves steadily. However, when the dimension of the hidden feature is larger than 120, the performance cannot always keep improving due to the larger computation burden.

Moreover, with regards to the number of deep layers, it is a known fact that more layers require more training data for the neural network to converge. Thus, the depth of the network is also determined by the amount of training data available, and this should always be considered when designing network architectures. We present the variation of the recognition results with different numbers of hidden layers in Fig. 6. Clearly, the CFCVAE achieves much better recognition performance but slower convergence speed with one hidden layer than that with two hidden layers. Thus, there is a tradeoff between the recognition performance and convergence speed.

According to the above analysis, in the following experiments, we adopt the neural network with one hidden layer and a feature layer with 120 Gaussian hidden units to attain the promising performance.

4.3. HRR radar target recognition performance

To illustrate the descriptive ability of the class-decoders in CFCVAE, we feed the test data into the trained model to generate the reconstructed samples. We randomly select a test sample from each class, and the reconstructed samples via the three class-decoders are presented in Fig. 7. Note that, in Fig. 7, different rows from top to bottom present the reconstructions of samples from An-26, Cessna Citation S/II, and Yark-42, respectively, and different columns from left to right show the reconstructions obtained by three class-decoders. For convenience, the decoders in CFCVAE are named class-decoder 1, class-decoder 2 and class-decoder 3, respectively. From Fig. 7, we can see that the class-decoder has much better descriptions for the test sample from the corresponding class, as presented in Figs. 7(a), (e), and (i). Furthermore, the root mean square error (RMSE) is adopted to give the quantitative results. The RMSE is defined as $\sqrt{\frac{1}{N_{test}}\sum_{n=1}^{N_{test}}\frac{\|\hat{\mathbf{x}}_n-\mathbf{x}_n\|_2^2}{\|\mathbf{x}_n\|_2^2}}$, where $\mathbf{x}_n$ denotes the nth original sample, $\hat{\mathbf{x}}_n$ represents the reconstructed sample of $\mathbf{x}_n$, and N_{test} denotes the number of test samples. The number of test samples from Yark-42, Cessna Citation S/II, An-26 is 1200, 2000, 2000, respectively. We list the RMSEs of the three class-decoders for the test samples from three classes in Table 2, from which the same conclusions as the Fig. 7 can be obtained.

By learning a specific decoder for the observed data from each class, the CFCVAE possesses several optimal class-decoders to depict the whole dataset precisely. Thus, it is reasonable to classify a sample to the class that corresponds to the class-decoder with the minimum reconstruction error. In the following, recognition experiments are performed to validate the superior performance of the proposed method, compared with some related models. The experimental results are presented in Table 3, where the results of PCA [3], K-SVD [4], AGC [17], CAGC [17], FA [17], CFA [17], SCAE [7], FDCVAE [9] come from their published papers. In addition, the SVAE is implemented by ourselves to show the performance of the traditional VAE incorporating the class labels. Compared with K-SVD [3] and PCA [4], the AGC [17] and FA [17] are the statistical models, which learn the probabilistic representations of the HRRP data and gain better performance. Besides, CAGC [17] and CFA [17] utilize the phase to further improve the performance of AGC and FA, respectively. In addition, the deep models could learn the features expressing the data better. The SCAE [7] contains multiple AEs to extract deep features of the HRRP and achieves the promising performance. Different from the SCAE, the SVAE and FDCVAE are the

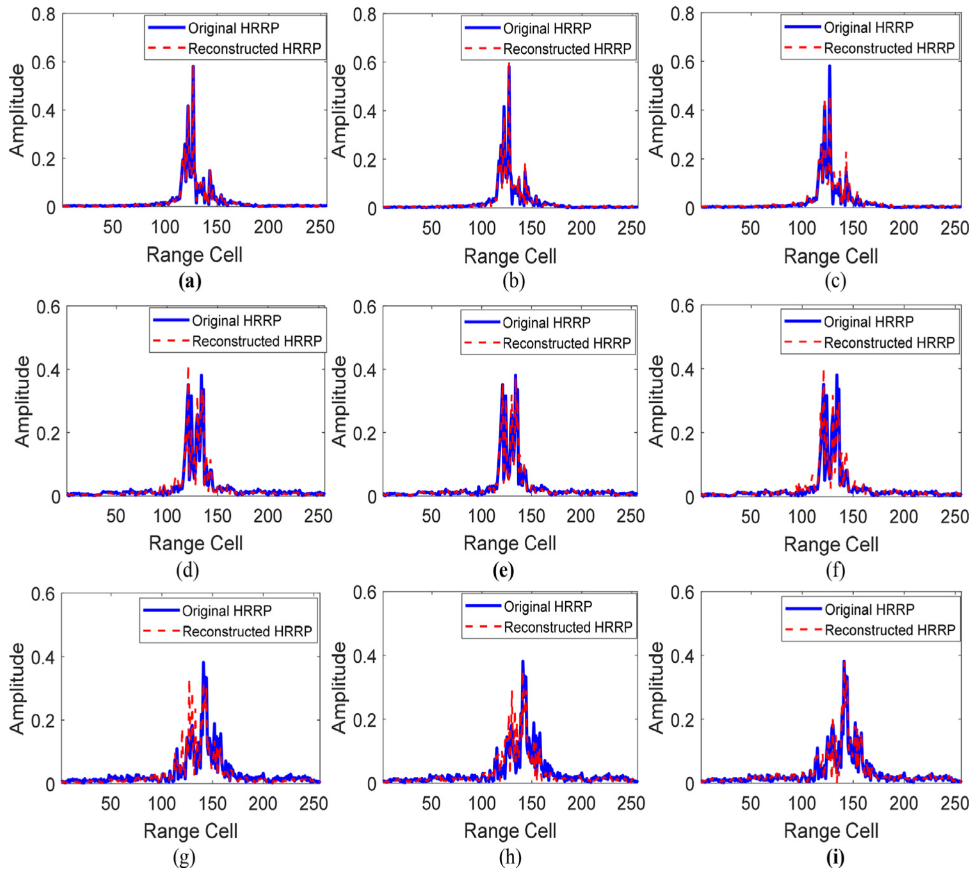

Fig. 7. Reconstructions of test samples via the class-decoders of CFCVAE. (a), (b) and (c) are the reconstructed HRRPs via the three class-decoders for the test sample from An-26; (d), (e) and (f) are reconstructed HRRPs via the three class-decoders for the test sample from Cessna Citation S/II; (g), (h) and (i) are reconstructed HRRPs via the three class-decoders for the test sample from Yak-42.

Table 2
Reconstruction error (RMSE) of the test data from different classes via the class-decoders of CFCVAE.

Test data/Class-decoders	Class-decoder 1	Class-decoder 2	Class-decoder 3
Test data from An-26	**0.1463**	0.3607	0.2914
Test data from Cessna Citation S/II	0.2358	**0.1453**	0.3208
Test data from Yak-42	0.2676	0.3465	**0.1708**

Table 3
Recognition performance of the CFCVAE and some related models.

Methods	Accuracy
PCA [4]	85.15
K-SVD [3]	85.32
AGC [17]	87.08
CAGC [17]	87.78
FA [17]	92.48
CFA [17]	92.56
SVAE	90.13
SCAE [7]	92.03
FDCVAE [9]	93.25
CFCVAE	**95.49**

deep generative models that learn the probabilistic features of the HRRP. And the FDCVAE [9] outperforms SVAE by leveraging the average profiles and conditional prior. Whereas for the CFCVAE, it takes advantage of the phase in HRR radar echo to attain the superior recognition performance based on the minimum reconstruction error criterion, which validates the effectiveness of our model.

The lack of training samples often results in overfitting, which greatly impacts the recognition performance. To illustrate the generalization ability of CFCVAE, in Fig. 8, we show its recognition performance varying with different numbers of training samples, as well as some related methods. For the sake of presenting the results clearly, we set the horizontal axis of Fig. 8 as the logarithm coordinate. In Fig. 8, except the results of PCA [4], K-SVD [3], AGC [17], CAGC [17], FA [17], CFA [17] and SCAE [7] models with 138,240 samples coming from their published papers, other results are implemented by ourselves. As we can see from Fig. 8, there is a clear trend that the recognition performance of these models is increasing with more training samples. For the AGC, CAGC, FA and CFA models, they need a large number of training samples to estimate the statistical parameters, especially for FA and CFA models. When the number of training samples is larger than 27,000, the performance of the FA and CFA models still improves greatly. And these statistical models attain much better performance than the PCA and K-SVD when training samples are sufficient. Moreover, the deep models outperform shallow models when the number of training samples is small. In particular, our CFCVAE always maintains much better performance than other methods under different

Table 4
Comparisons of recognition performance for the CFRVAE and CFCVAE with different numbers of training samples.

Methods/ The number of training samples	540	1350	2700	13,500	27,000	138,240
CFRVAE	86.22	87.82	88.12	88.37	90.01	90.04
CFCVAE	87.56	90.02	92.43	95.17	95.19	95.49

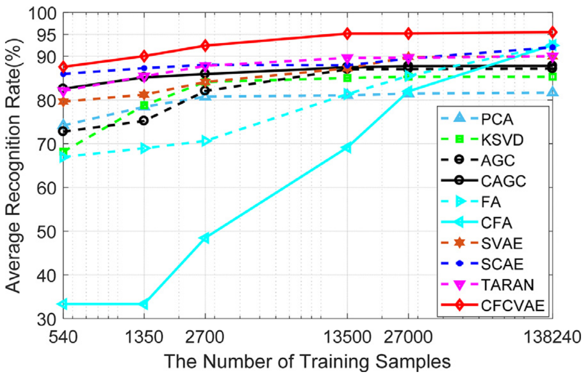

Fig. 8. Recognition performance of the CFCVAE and other related models with different numbers of training samples. The results of the PCA [4], K-SVD [3], AGC [17], CAGC [17], FA [17], CFA [17], SCAE [7] with 138,240 training samples come from the published papers.

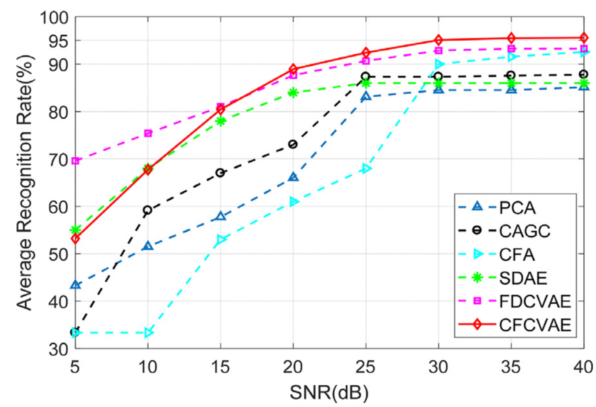

Fig. 9. Recognition performance of the CFCVAE and some related models under different SNRs. The results of PCA [1], CAGC [17], CFA [17], SDAE [12], FDCVAE [9] come from the published papers.

numbers of training samples. In CFCVAE, multiple class-decoders have the capacity to give a more accurate depiction for the observations, which promotes the generalization ability of our model.

Furthermore, we investigate the effect of the phase in HRR radar echo on our CFCVAE. Corresponding to the CFCVAE, a real domain model, named class factorized real variational auto-encoder (CFRVAE), is constructed for radar HRRP recognition. In Table 4, we list the recognition results of the CFCVAE and CFRVAE with different numbers of training samples. As we can see from Table 4, the recognition performance of the two models both improves when the number of training samples increases. Moreover, the CFCVAE always outperforms CFRVAE, and the gap widens along with the increase of training samples. To sum up, the phase in HRR radar echo is useful for recognition and contributes to the excellent performance of CFCVAE.

4.4. Noise robustness performance

In the following, we analyze the noise robustness performance of the CFCVAE, compared with some methods including PCA [1], CAGC [17], CFA [17], SDAE [12], FDCVAE [9], with test data under different signal-to-noise ratio (SNR) conditions.

The measured data used in this paper is obtained under high SNR (≈40 dB). In order to evaluate the noise robustness of our method, we add simulated white noises to the complex test samples. The definition of SNR [26] is

$$SNR = 10 \times \log_{10}\left(\frac{\sum_{l=1}^{L} P_{\mathbf{x}_l}}{L \times P_{noise}}\right), \tag{28}$$

where $P_{\mathbf{x}_l}$ represents the signal power of the lth ($l \in \{1, \cdots, L\}$ with L denoting the number of the range cell) range cell from the original test samples, and P_{noise} denotes the power of noise in each range cell. For the measured dataset, $L = 256$.

We show the results of our method and some models in Fig. 9, in which the results of PCA [1], CAGC [17], CFA [17], SDAE [12], and FDCVAE [9] come from their published papers. As shown in Fig. 9, with the increase of noise, there is a clear decreasing trend in the recognition performance of all methods. Since the nonlinear network is capable of learning robust features for recognition, the proposed CFCVAE gains better performance than the shallow models including PCA [1], CAGC [17] and CFA [17]. When $SNR \geq 15dB$, the CFCVAE is superior to SDAE [12]. However, by reconstructing the original data from the noised data, the SDAE [12] learns the noise-robust features so as to achieve a litter better performance than CFCVAE when $SNR \leq 10dB$. In addition, with the help of average profiles and Batch-Normalization operation, the FDCVAE [9] performs better than CFCVAE when $SNR \leq 15dB$, but when $SNR \geq 20dB$, the CFCVAE outperforms FDCVAE [9]. Therefore, the CFCVAE achieves much better performance than most of the contrastive methods under different SNRs, which suggests that our model is robust to the noise.

4.5. Computation burden

In real applications, for an off-line learning (or training) model, the computation burden in the test stage is significant to evaluate the practicability of the model. In the second column of Table 5, we present the computation burden of the test stage for the proposed CFCVAE and some related models. In detail, based on the analysis in Section 4.2, the CFCVAE contains a hidden layer with L_1 hidden units, and the dimension of the feature is L_2. In the test stage of the CFCVAE, a sample is classified by a rapid means of feed-forward propagation via the trained network and the computation burden is proportional to P. As for PCA, AGC, FA, CAGC and CFA models, they calculate the conditional likelihood with all the target-aspect templates and then search the maximum one. In this way, the computation burden of these models is quadratic to P, which means these methods are usually time-consuming. While for K-SVD algorithm, it uses the OMP [20] to reconstruct the test sample with the dictionaries learned by the training data from different classes, and then classifies the sample to the class with corresponding dictionary having minimum reconstruction error. In our experiments, the iteration number K of the OMP is usually approximated to $\frac{P}{2}$. The computation burden of the K-SVD is four power to the dimension P, which shows the method is much more time-consuming. According to the above analysis, the CFCVAE has much less computation burden than related methods. Moreover, in the last column of Table 5, we show the time cost of all test samples,

Table 5
Computation burden and test time of the CFCVAE and some related models.

Method	Computation burden in test stage	Test all samples (s)
PCA	$O(N_{test}P^2 \sum_{c=1}^{C} M_c)$	25.5321
AGC		21.2246
FA	M_c denotes the number of frames in cth class.	25.7473
CAGC		28.4372
CFA		38.7854
K-SVD	$O(N_{test}(\sum_{k=1}^{K}(k^2P + kP + k^3 + k^2) + P))$	122.9512
	K denotes the number of iterations in OMP.	
CFCVAE	$O(N_{test}(P \times L_1 + L_1 \times L_2 + \sum_{c=1}^{C} L_1 \times (L_2 + P)))$	4.1604

which illustrates that our method consumes much less time at testing. Thus, the proposed model has much higher time-efficiency than these related models.

5. Conclusion

This paper presents a novel generative framework named CFCVAE for HRR radar target recognition. To explore the deep probability representation of the HRR radar echo, the proposed CFCVAE is constructed in a CV-NN framework. Moreover, the observed data are divided into several parts based on the class labels, and each part is described by a specific class-decoder. Due to the stronger description capability of the single decoder for a part of the observed data, the CFCVAE can give a more precise depiction to the whole dataset. In addition, our model adopts the conditional prior based on the class labels to improve the discriminative ability of the latent features. The proposed CFCVAE is a discriminative model in which a sample is predicted to the class corresponding to the class-decoder with the minimum reconstruction error. The phase in HRR radar echo, multiple class-decoders and the conditional prior all contribute to the excellent recognition performance of the CFCVAE. Moreover, our method also shows the strong performance in terms of noise robustness and computation burden.

In the future, we are interested in improving the noise robustness performance of the proposed model. Moreover, we are also interested in extending the CFCVAE to a complex-valued convolutional neural network, which can be utilized for the classification of the complex-valued image.

Declaration of Competing Interest

None.

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